proposal

Graded

Group

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...and 1 more
View or edit group

Total Points

36 / 40 pts

Question 1

Proposal

36 / 40 pts

introduction

- **– 1 pt** partially complete

objectives and specifications

- **– 1 pt** mostly complete

project strategy

- **– 1 pt** mostly complete

tasks and priorities

- **– 0 pts** complete

research plans

- **– 1 pt** mostly complete

schedule

- **– 0 pts** complete

bill of materials

- **– 0 pts** complete

1. could use additional details about the user interface
2. what accuracy is expected?
3. this duration may limit functionality

Question assigned to the following page: 1

ECE436 IoT Project Proposal

ESP32 LoRa GPS Pet Tracking and Behavior Monitoring System

Bryce Huey, Ely McLemore, Owen Macrowski, and Tanner Staus

Introduction

For our project, we are planning on tracking a dog wearing a GPS tracker collar, with a website interfacing with a server to show their current and previous locations on a satellite map. Our project intends to show off a fun and practical way to use GPS technology with people's pets, observe possible movement or behavior patterns, and locate their pet wherever they might be.

Objectives & Specifications

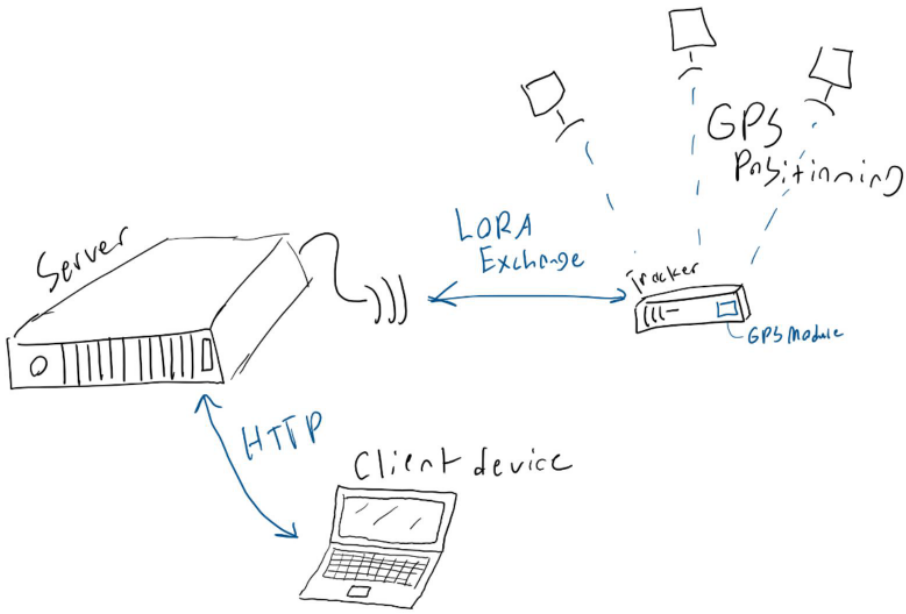
Our project aims to track dogs in a way that allows users to be able to know where their pet is always though a web interface. Users will be able to start and stop tracking their dog tags by connecting over Wi-Fi to the server's HTML. The server will be a high-power device while the trackers will be low power optimized. Sleep functionally will be implemented into the trackers to allow for a longer battery life.

- Dog trackers can actively run for 2 hours on tracking mode.
- The trackers can operate at least 100 meters from the server.
- Tracker modules can be used in any part of the world (not region specific).
- Server website can be accessed by a range of devices over a local network.
- Trackers will allow the ability to recharge in a fashion that does not require excessive work.
- Server will at least support 4 running trackers at once.

Question assigned to the following page: 1

Project Strategy

Initially, we will design a prototype for all 4 main parts of our project. This will include figuring out how to get the LoRa radio communication working as well as figuring out its practical range and data transmission speeds/size. We also need to design a basic webpage that stores the data from the LoRa communication and displays it for the users. Additionally, we are going to design an enclosure for the LoRa radio, GPS, and IMU modules that will make them secure, less prone to the elements, and able to be attached to a dog's collar. Lastly, we also need to get the IMUs calibrated and working so we can use them to detect different behavior of the dog. As the project progresses we aim to build on top of these core features and potentially add additional ones as well as improving the user interface and experience.



Question assigned to the following page: 1

Research Plans

- Design a mock UI for the web interface
- Figure out our preferred GPS components
- Design a flowchart for the communication framework
- Auxiliary methods of connecting to the GPS collar unit
- Exploring the effective range of the GPS devices in person
- Efficient website/server creation techniques, if possible

Tasks & Priorities

- Setup
  - Create a GitHub repository for source control
  - Order our preferred components
- Frontend
  - Displays data in a useful matter
  - Have good overall user experience
  - All for user input (turn on and off trackers)
- ESP
  - Position localization (tracking) using gps module
  - Reporting data to server/webpage
  - Operate on battery with sleep and awake modes
- Backend
  - Stores all data collected in a file it can be read by frontend
  - Manage connected device tracking
  - Allow for remote control of devices
    - Uses user input from front end
    - Turning device low power modes on/off
    - Setting polling rates on the fly

Roles

Bryce – Design box and get GPS positioning to work

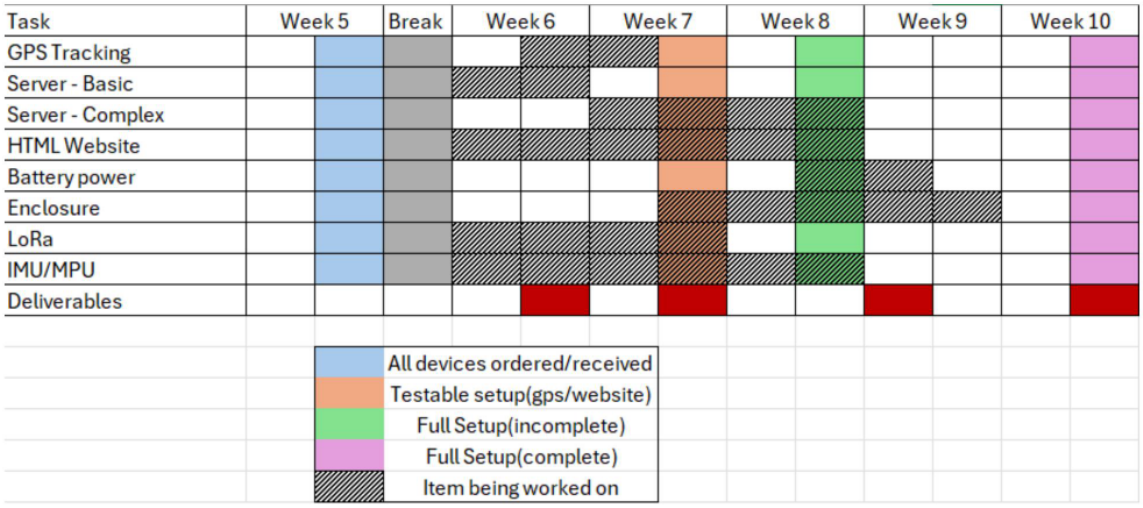
Ely – Get LoRa Radio to work to send position and IMU data to Pi server

Owen – HTML web interface on the Pi server

Tanner – Get IMU to figure out dog behavior/activity

Question assigned to the following page: 1

Schedule



Budget

Qty	Part	Description	Notes	Cost
4	L76k GPS for ESP 32	GPS interface for the esp32 board	Ordered	$15.08
4	Heltec Lora V4 kit	Esp32 board and battery	Already Owned	$148.00
1	Heltec LoRa V3	Esp32	Already Owned	$20.00
4	GY-521	Inertial measuring unit	Already Owned	$4.00
				Total: $187.08